LECTURE 15 · DISCRETE MATHEMATICS

Repair relations. Discover classes.

Which missing connections are forced?



A closure adds only forced pairs

For reflexivity, symmetry, or transitivity, the closure is the smallest relation containing R with the chosen property.

Keep every original pair

$$R \subseteq C$$

Make the rule hold

Add every pair required by the rule.

Add nothing unnecessary

If $R \subseteq S$ and S has the property, then $C \subseteq S$.

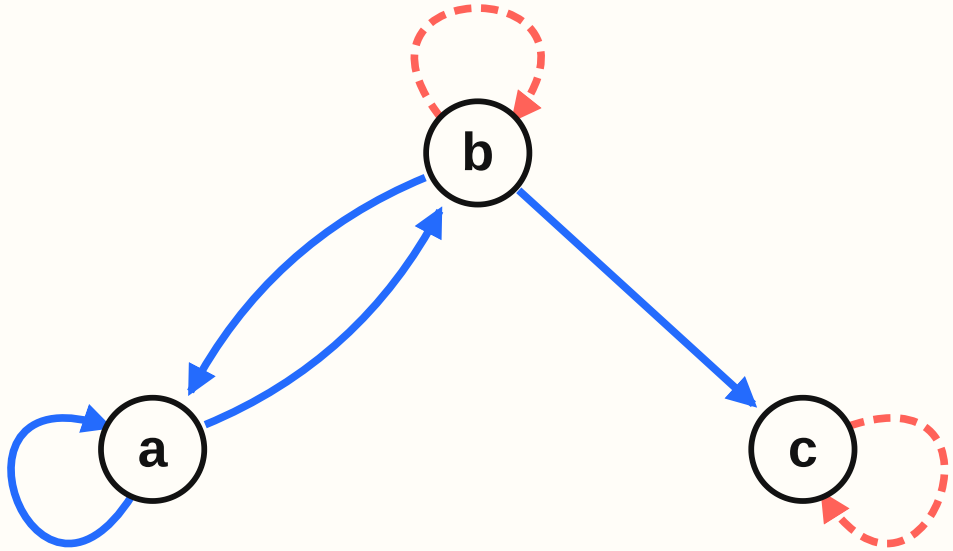
“Smallest” means contained in every relation that meets the requirements.

EXERCISE 1

Reflexive closure: add the missing loops

$A = \{a, b, c\}, \quad R = \{(a, a), (a, b), (b, a), (b, c)\}$. Keep all four pairs.

01 **Which pairs must be added?**
(b, b) **and** (c, c)**.**



Every element of the underlying set needs a loop.

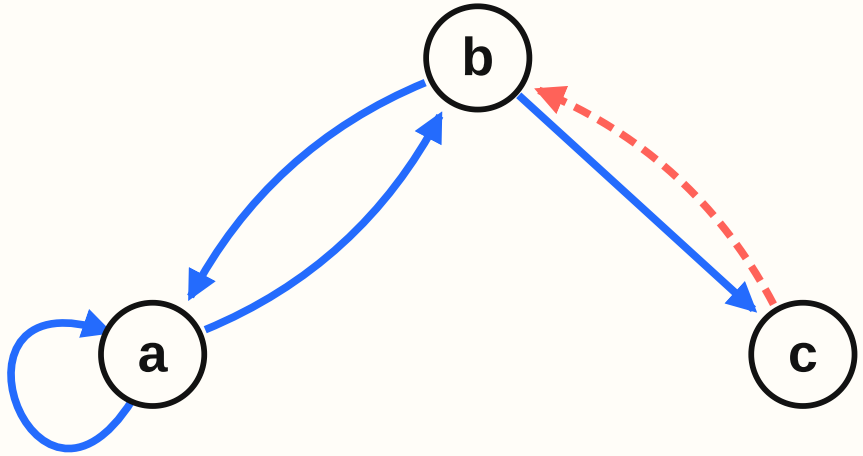
EXERCISE 2

Symmetric closure: add reverse arrows

$$R = \{(a,a), (a,b), (b,a), (b,c)\}.$$

01 **Which pair must be added?**

(c,b), **the reverse of** (b,c)**.**



An existing reverse arrow does not need to be added twice.

Two closures have direct formulas

Reflexive closure

$$r(R) = R \cup \Delta_A$$

$$\Delta_A = \{(a, a) : a \in A\}.$$

Symmetric closure

$$s(R) = R \cup R^{-1}$$

$$R^{-1} = \{(b, a) : (a, b) \in R\}.$$

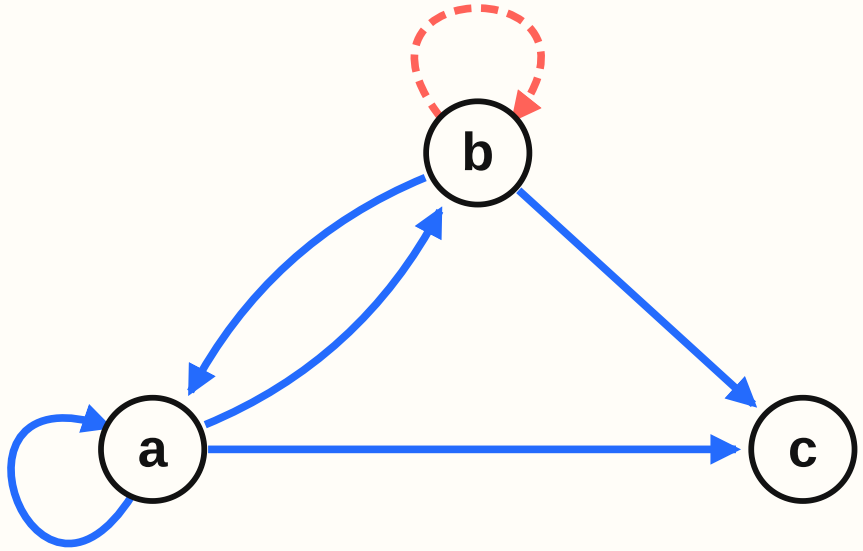
For each formula: containment, the required property, and minimality.

EXERCISE 3

A shortcut is not the whole repair

Add (a, c) because aRb and bRc. Is the result transitive?

01 **Which other pair is forced?**
(b, b)**, because** bRa **and** aRb**.**



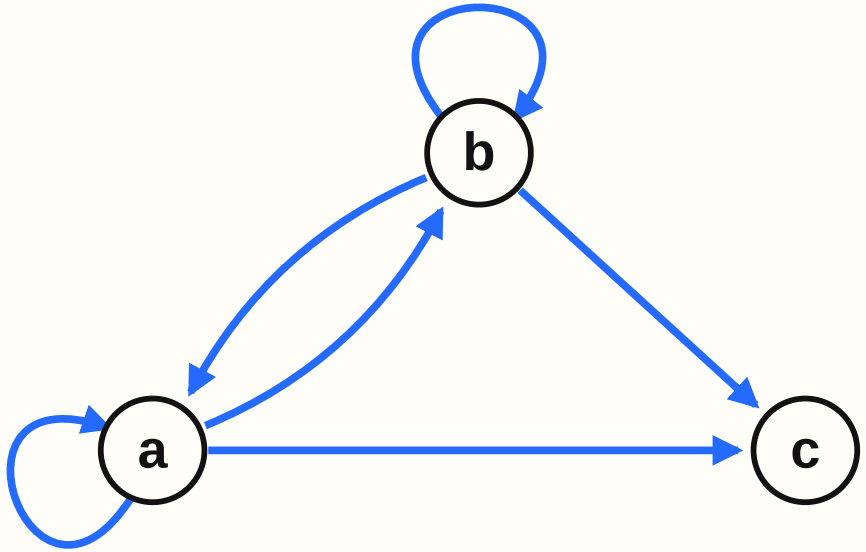
A two-edge path may begin and end at the same vertex.

The transitive closure is now complete

$$t(R) = R \cup \{(a, c), (b, b)\}.$$

From a: reach a, b, c. From b: reach a, b, c.

From c: no outgoing path, so no new pair.



Every two-step route already has its shortcut.

Transitive closure records reachability

$(x, y) \in t(R)$ exactly when there is a path from x to y using one or more edges of R.

Contains R

An original edge is a path of length one.

Transitive

Concatenate a path from x to y with one from y to z.

Smallest

Every transitive relation containing R must contain every path shortcut.

A loop is forced by a positive-length cycle, not by an empty path.

EXERCISE 4

One pass may miss a longer route

$$A = \{1, 2, 3, 4\}, \quad R = \{(1, 2), (2, 3), (3, 4)\}.$$

01 **Find $t(R) \setminus R$.**

$\{(1, 3), (2, 4), (1, 4)\}$.

02 **Why must (1,4) appear?**

It is forced by the path $1 \to 2 \to 3 \to 4$.

Continue until adding shortcuts creates no new obligations.

EXERCISE 5

When do symmetry and transitivity force loops?

Assume $\forall a \in A\ \exists b \in A :\ aRb$ or bRa.

01 **Must $t(s(R))$ be reflexive on A?**

Yes. Symmetry supplies both directions, then transitivity supplies each loop.

Use the incident edge in either direction

Fix an arbitrary element

Choose $c \in A$. There is $d \in A$ with cRd or dRc.

After symmetric closure

Both (c, d) and (d, c) belong to $s(R)$.

After transitive closure

$(c, c) \in t(s(R))$. This holds for every $c \in A$.

The same argument works when c = d.

EXERCISE 6

An isolated element changes the answer

$$A = \{a, b, c\}, \quad R = \{(a, b)\}.$$

01 **Find $t(s(R))$.**

$\{(a, a), (a, b), (b, a), (b, b)\}$.

02 **Why is it not reflexive on A?**

(c, c) is missing: c has no incident edge.

Symmetric and transitive does not imply reflexive on the whole set.

Three repairs, three obligations

Closure	What must be added?	What may remain absent?
Reflexive	Every missing loop	Reverse arrows
Symmetric	Every missing reverse arrow	Loops at isolated elements
Transitive	Every path shortcut	Loops without a return path

Always name the underlying set before computing a closure.

MODULAR ARITHMETIC

Arithmetic that cycles

Different integers can represent the same remainder.



Congruence means the same remainder

Fix a positive integer m. Integers a, b are congruent modulo m when division by m gives the same remainder.

$$a \equiv b \pmod{m} \iff m \mid (a - b)$$

Example

$$12 \equiv 2 \pmod{5}$$

$12 - 2 = 2 \cdot 5.$

Negative integers belong too

$$-3 \equiv 2 \pmod{5}$$

$-3 = (-1) \cdot 5 + 2.$

Add, subtract, multiply, then reduce

Operation	Example modulo 5
Addition	$7 + 5 = 12 \equiv 2 \pmod 5$
Subtraction	$11 - 8 = 3 \equiv 3 \pmod 5$
Multiplication	$6 \cdot 4 = 24 \equiv 4 \pmod 5$

You may replace operands by congruent integers before these operations.

EXERCISE 7

Compute the remainders

Give the representative in $\{0, 1, 2, 3, 4\}$.

01 $18 + 14 \pmod 5$

$32 \equiv 2 \pmod 5$.

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02 $2 - 9 \pmod 5$

$-7 \equiv 3 \pmod 5$.

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03 $13 \cdot 12 \pmod 5$

$3 \cdot 2 = 6 \equiv 1 \pmod 5$.

Division requires a multiplicative inverse

An inverse of b modulo m is an integer u with $bu \equiv 1 \pmod{m}$.

Find the inverse

$$3 \cdot 5 = 15 \equiv 1 \pmod{7}$$

Solve the division problem

$$3x \equiv 12 \pmod{7} \quad \Longrightarrow \quad x \equiv 5 \cdot 12 \equiv 4 \pmod{7}$$

For $m > 1$, an inverse exists exactly when $\gcd(b, m) = 1$.

EXERCISE 8

Ordinary cancellation can fail

Work modulo 6.

01 **Does 2 have a multiplicative inverse?**

No: $2u$ has remainder 0, 2, or 4, never 1.

02 **Solve $2x \equiv 2 \pmod 6$ for $0 \le x < 6$.**

$x = 1$ or $x = 4$. Cancelling 2 would lose a solution.

Multiply by an inverse only after establishing that it exists.

EQUIVALENCE RELATIONS

Relations form classes

What makes “belongs to the same group” precise?



An equivalence relation satisfies all three

Reflexive

aRa **for every** $a \in A$**.**

Symmetric

$aRb \implies bRa$**.**

Transitive

$aRb \wedge bRc \implies aRc$**.**

Each element relates to itself and to every member of its class.

Congruence is an equivalence relation

On $\mathbb{Z}$, define $aRb \iff m \mid (a-b)$.

Reflexive

$$a - a = 0 = m \cdot 0.$$

Symmetric

$$a - b = mk \Longrightarrow b - a = m(-k).$$

Transitive

$$a - b = mk,\ b - c = m\ell \Longrightarrow a - c = m(k + \ell).$$

Each property follows from an integer multiple of m.

A class contains all equivalent elements

$$[a] = \{x \in A : xRa\}$$

Modulo 5

$$[2] = \{\ldots, -8, -3, 2, 7, 12, \ldots\}$$

Choose another representative

$$[2] = [7] = [-3]$$

The representative names the class; it is not the only element in it.

Modulo five gives five classes

Class	All members
$[0]$	$\{5k : k \in \mathbb{Z}\}$
$[1]$	$\{5k + 1 : k \in \mathbb{Z}\}$
$[2]$	$\{5k + 2 : k \in \mathbb{Z}\}$
$[3]$	$\{5k + 3 : k \in \mathbb{Z}\}$
$[4]$	$\{5k + 4 : k \in \mathbb{Z}\}$

Every integer belongs to exactly one row.

EXERCISE 9

Class names can differ without changing the class

Use congruence modulo 5 on $\mathbb{Z}$.

01 **Which of $[-8], [2], [7], [12], [3]$ are equal?**

$[-8] = [2] = [7] = [12]$. **The class $[3]$ is different.**

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02 **Find $[2] \cap [3]$ and $[2] \cup [7]$.**

$[2] \cap [3] = \varnothing$ **and** $[2] \cup [7] = [2]$.

A partition covers without overlap

A partition of A is a family of subsets of A satisfying:

Nonempty parts

$B \neq \varnothing$ **for every part** B**.**

Coverage

$\bigcup B = A$ **(union over all parts).**

Disjointness

$B \neq C \Longrightarrow B \cap C = \varnothing$**.**

A partition can have finitely or infinitely many parts.